

Does the presence of leads in the Arctic sea ice have an impact on the ocean?

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Abstract

The Arctic sea-ice, in particular the ice-pack, acts as an insulator between the atmosphere and the ocean below. Leads, common in Arctic sea ice, facilitate exchanges of heat, salt, moisture and gases between the ocean and the atmosphere. Therefore, leads can influence heat exchanges, ice melt or growth, ocean dynamics, and the atmospheric boundary layer. The existing literature suggests that leads play a significant role in the heat budget of the surface of the ocean and likely impact water mass transformation within the Arctic. However, due to the lack of observations and high resolution models capable to resolve leads, an estimate of the impact of leads in the ocean has been elusive. Here, we explore the magnitude and impact in the ocean surface caused by the leads using SEDNA, a really high-resolution pan-Arctic hindcast (1/60°; NEMO + SI3 configuration). Leads account for **20%** of the sea-ice cover area, and within them, there is approximately 20% of the total surface water mass transformation. The buoyancy flux and water mass transformation in leads are similar to those underneath the surrounding floes and the percentage of water mass transformation through leads matches their area coverage. Thus, the present estimate indicates that leads have a small contribution to Arctic Ocean dynamics, contrary to previous hypotheses.

Keywords: Arctic, Leads, Buoyancy flux

1 Introduction

The Arctic sea-ice plays a crucial role in regulating Earth’s climate by acting as a natural insulator between the atmosphere and the ocean below (Wettlaufer et al, 1997; Untersteiner, 1961). This insulating property acts as a thermal barrier, limiting heat loss from the ocean to the atmosphere, minimizing evaporation, and maintaining the stability of the polar environment (Ruffieux et al, 1995). A ubiquitous feature of Arctic sea ice is the formation and persistence of leads. Leads are regions where the sea ice deformation reaches a local maxima (See methods for the definition of sea ice deformation; Hutter et al 2019). These features are characterized by sea ice shear and divergence that can result in an opening of the sea ice, and thus allow large flux exchanges between the ocean and the atmosphere (Lüpkes et al, 2008). Leads vary in width from a few meters to kilometers, with length-scale spanning distances of a few kilometers up to hundreds of kilometers, and generally last from a few hours up to a few days (Linow and Dierking, 2017; Wernecke and Kaleschke, 2015; Tschudi et al, 1998). Leads can be formed through the divergence or/and shear of the sea ice forced by winds and oceanic currents (Rampal et al, 2016; Hutchings et al, 2005; Richter-Menge et al, 2002). Winds are thought to be the main forcing of the sea ice deformation and the formation of leads (Linow and Dierking, 2017; Hutter et al, 2018). Meanwhile, ocean currents have a smaller contribution to the sea ice deformation compared to wind forcing, but the ocean drag at the bottom of the ice acts to balance the wind-induced sea ice velocities, stabilizing the sea ice cover (Weiss and Marsan, 2004). Additionally, observations show that ocean currents may play a bigger role in dictating the fracturing patterns of sea ice, since regions with strong currents and bathymetric features are collocated with a larger occurrence of leads (Willmes et al, 2023).

Once leads are formed, atmospheric forcing in conjunction with localized upwelling can result in important heat fluxes at the ocean surface ($\sim 400W/m^2$; McPhee et al, 2005; Marcq and Weiss, 2012; Maykut, 1986), instigating localized melting/freezing of sea ice (von Albedyll et al, 2022). Additionally, changes in buoyancy forcing due to brine rejection into the mixed layer can induce convection (Smith and Morison, 1998), leading to the generation of fronts, mixed layer turbulence, and eddy generation (Reiser et al, 2020; Smith et al, 2002). Furthermore, changes in the buoyancy flux could result in the local transformation the surface water masses (Pemberton et al, 2015; Smith et al, 1990; Walin, 1982). Although all the previous studies have shown that instances of leads can locally impact the ocean dynamics, the overall impact of leads to the ocean dynamics across the Arctic basin are yet to be quantified.

In-situ observations have provided evidence that leads locally impact the ocean structure of the Arctic Ocean. Nevertheless, the lack of observations limits an estimation of their integrated impact on the Arctic Ocean. Here, we present the first estimate of the impact of the leads in the Arctic Ocean, using a really high-resolution pan-Arctic hindcast, SEDNA (Talandier and Lique, 2023). Since the ice cover is dynamically and thermodynamically different in the marginal ice zone (MIZ) and the ice pack, we differentiate between the leads in each region. Hereafter, we refer to leads as the union of the MIZ leads and the ice pack leads or LKFs. The impact of MIZ leads and ice pack leads is quantified by using buoyancy fluxes and the water mass transformation framework (See Methods for details of the calculations and numerical model).

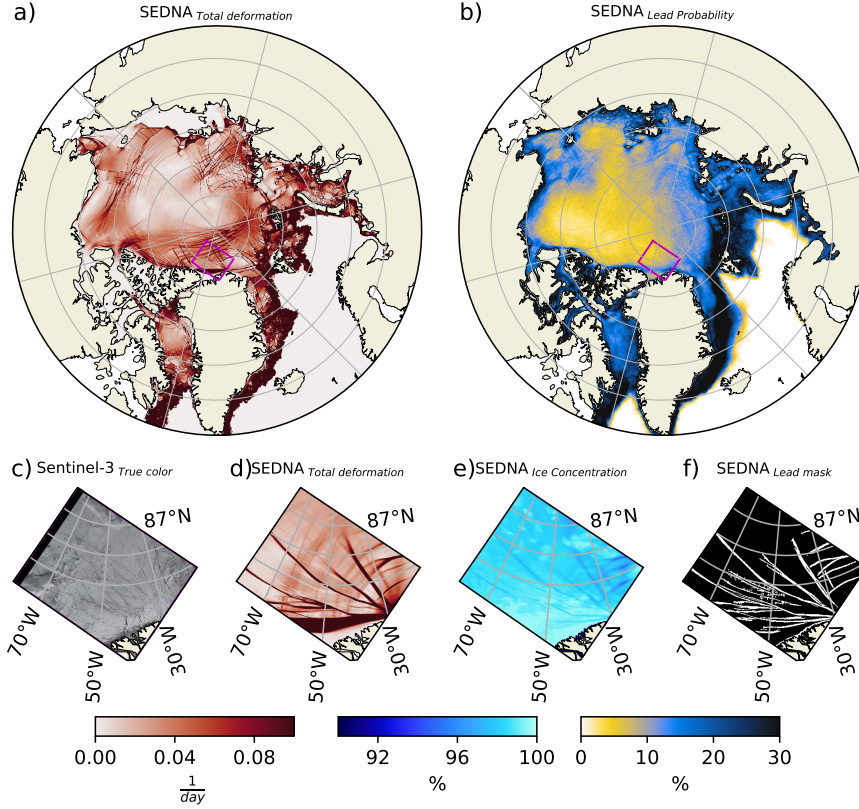


Fig. 1 a) Snapshot of the total deformation for the 21st of April 2014. b) Mean lead frequency between 2011 and 2015. c) True color image from Sentinel 3 on the 12th of April 2022 for the magenta area in panel a. d) Snapshot of the ice concentration from SEDNA on the 21st of April 2014. e) Total deformation used to identify leads. f) Mask of identified leads on the 21st of April 2014.

2 Lead contribution to the ocean buoyancy

Climate models and coupled models with resolutions lower than $4.5km$ are unable to reproduce many aspects of leads due to the lack of resolution to capture their spatial scale (Wang et al, 2016). SEDNA is capable of overcoming this limitation and capture most of the lead width distribution ($> 1km$; Wernecke and Kaleschke 2015), since the model resolution is $\sim 800m$ on average across the Arctic. A snapshot of the total deformation is shown in figure 1a. A closer examination, focusing north of Greenland in panels c, d, e, and f of figure 1, reveals similarities between the spatial pattern of typically observed leads on the 12th of April 2022 from a true color satellite image (Fig. 1c) and the ice concentration of SEDNA for the 21st of April 2014. Note that the sea ice concentration in the identified leads shown in figure 1f using the lead identification algorithm proposed by Hutter et al (2019) remains higher than 90% concentration; this is a consequence of the visco-plastic nature of the sea ice model and the daily averaged output that smooths the presence of leads in the fields. Despite this, the

139 similarities between the observations and model lead probability allow us for the first
140 time to estimate quantitatively the impact leads have on the Arctic ocean.

141 The climatology of the lead probability, i.e. the likelihood of a leads occurring for
142 any given day, is shown in figure 1b. The mean probability of leads averaged over
143 the Arctic is $\sim 19\%$ in SEDNA, comparable to observations using RADARSAT Geo-
144 physical Processor System and Thermal-Infrared Satellite Imagery (Wernecke and
145 Kaleschke, 2015; Willmes et al, 2023). Although observations use different identifica-
146 tion methods compared to the one implemented in SEDNA (Hutter et al, 2019), there
147 is resemblance between the spatial patterns of SEDNA and observations (Fig. S1).
148 For example, we observe increased lead frequency in the shelves, along bathymetric
149 features, and in regions with large kinetic energy (Fig. S1). Several hotspots of lead
150 occurrence can be identified in the Chukchi Sea, highlighting the model’s ability to
151 reproduce leads dynamics. SEDNA reproduces the patterns observed in the Arctic
152 shelves, the Greenland Sea, Barents Sea, Kara Sea and Laptev Sea, but the prominent
153 signature in the Beaufort Sea in observations is not captured in SEDNA.

154 Figure 2a and b show two snapshots of the buoyancy flux. In winter, a negative
155 buoyancy flux results from sea ice refreezing and brine rejection. In summer, a positive
156 buoyancy flux is associated with sea ice melting and freshening of the Arctic surface
157 (Fig. 2b). Inside the ice covered region (magenta contour) in figure 2, the fluxes are
158 spatially homogeneous, except for the marginal ice zone and the leads, where the
159 buoyancy fluxes are larger than those underneath the surrounding floes, in agreement
160 with previous studies. The flux across the Arctic for the leads is $-2.46 \times 10^{-8} m^2/s^3$
161 and $7.20 \times 10^{-8} m^2/s^3$ for each date, respectively, compared to the $-2.42 \times 10^{-8} m^2/s^3$
162 and $5.89 \times 10^{-8} m^2/s^3$ underneath the surrounding floes. In other words, in winter, the
163 fluxes across the leads and the surrounding floes are comparable. However, in summer,
164 the fluxes in the leads can be up to 20% larger than those underneath floes.

165 Two transects across the Beaufort Gyre are shown in figure 2 for the 1st of January
166 2014 and 25th of June 2014. Examining the 1st of January 2014, we observe a small
167 heat flux in the leads (Fig. 2 b), which does not impact the heat flux at the ocean-
168 atmosphere interface since the ocean surface is at freezing point. In other words, the
169 extra heat exchanged with the atmosphere is used to form sea ice instead of further
170 cooling the ocean, resulting in a heat flux close to zero at the ocean-ice-atmosphere
171 interface. In contrast, the salt flux (Fig. 2e) shows two positive peaks, each with twice
172 the magnitude of the surrounding floes, collocated within the identified leads. This is
173 due to an enhanced sea ice growth and brine rejection within the leads. The salt flux
174 and heat flux result in a negative buoyancy flux for this snapshot where the largest
175 buoyancy loss occurs within the leads (Fig. 2 j). Additionally, within the leads there
176 is a decrease in sea ice thickness of 25cm (Fig. 2 g). On the 25th of June (Fig. 2d), we
177 observe an increase in the heat fluxes within some of the identified leads, reaching up
178 to $\sim 25 W/m^2$. Meanwhile, the freshwater flux shows a local decrease within the leads,
179 indicating a freshening of the surface due to sea ice melt. This change is captured
180 by the positive buoyancy flux, decreasing the density at the surface (Fig. 2k). The
181 enhanced sea ice melt in the leads results in a sea ice thickness decrease of more than
182 1m within the leads (Fig. 2 h). Note that some peaks in the buoyancy flux align with
183 the edge of the detected leads, likely due to the enhanced refreezing or melting that
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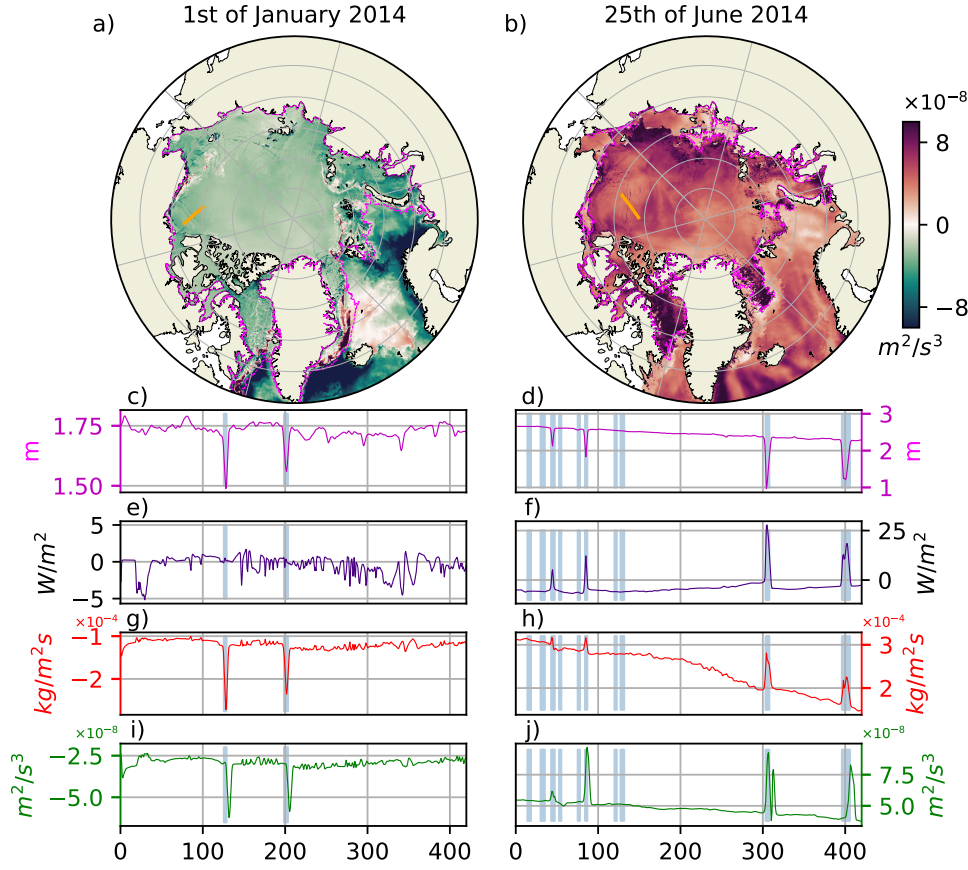


Fig. 2 Maps of buoyancy flux and transects for the 1st of January and 25th of June 2014. a) Map of buoyancy flux on the 1st of January 2014. Transect on the 1st of January 2014 of c) downward heat flux, e) freshwater flux, g) ice thickness and i) buoyancy flux. b) Map of buoyancy flux on the 25th of June 2014. Transect on the 25th of June 2014 of d) downward heat flux, f) freshwater flux, h) ice thickness and j) buoyancy flux. Note that the transect location and orientation changes between the winter and summer dates. Magenta contour in panels a and b correspond to the 0% concentration contour of sea ice. Vertical blue bars shows the mask of leads in the transect.

occur at the edge of the lead. Of all the identified leads, some exhibit significant ocean-atmosphere fluxes, while others have little to no imprint in the fluxes. Leads without large ocean-atmosphere fluxes may be explained by the temporality of the sea ice break up (divergence), where later snapshots show an increase of fluxes at the location of identified leads. Alternatively, lead generated through shear and convergence have a limited flux exchange with the atmosphere, but may have enhanced fluxes in the ocean interior due to isopycnal upwelling forced by the ice-ocean stress (Bourgault et al, 2020).

Similar to observations, we identify fluxes that are larger in the leads than in the ice pack floes in both summer and winter. However, the freshwater flux, attributable

to melting and freezing, is significantly larger than the heat fluxes associated with the leads. In the next section, we explore the impact that leads have in the magnitude of the buoyancy fluxes and water mass transformation across the Arctic basin during 2014. This analysis is conducted separately for the marginal ice zone (MIZ; 0-80% sea ice concentration), ice pack (80-100% sea ice concentration) and the leads within each of these regions.

3 Mean contribution of leads in the Arctic

In the Arctic ice covered ocean, the averaged percentage area covered by the four distinct masks representative of the sea ice covered Arctic Ocean during 2014 is as follows: 19% for the MIZ, 81% for the ice pack, 14% for the LKFs in the ice pack, and 10% for the leads in the MIZ. The area covered by each mask exhibit a seasonal cycle (Fig. 3a). During summer (June, July, and August), the percentage of the ice pack cover area decreases from approximately 90% to 50%, while the MIZ increases from 10% to approximately 50%. On one hand, the percentage of the area covered by LKFs in the ice pack peaks in winter (December, January, and February) at almost 20% coverage over the ice pack area. On the other hand, the leads in the MIZ peak in summer, covering up to 25% of the total ice cover Arctic and 50% of the MIZ surface area. The maxima leads area coverage is $\sim 20\%$ of the total sea ice covered Arctic, consistent with previous model assessments (Wang et al, 2016).

To quantify the contributions of leads to ocean buoyancy, we integrate the buoyancy flux over each of the sea ice cover masks (Fig. 3b), where the extent of each ice cover mask evolves temporally following the seasonal cycle of the sea ice extent. The buoyancy flux shows a significant seasonal cycle, with positive values corresponding to the sea ice melting season and negative values coinciding with the sea ice freezing season. The mean buoyancy flux of the ice pack in winter (December-February) and summer (June-August) is $-0.32m^2/s^3$ and $0.34 \times 10^{-14}m^2/s^3$, respectively. The buoyancy flux in the ice pack indicates a shift from ice formation in winter to ice melting in summer, dominated by the haline buoyancy component. Meanwhile, the mean buoyancy flux in the MIZ in winter and summer is $0.01m^2/s^3$ and $0.25m^2/s^3$, respectively. A positive buoyancy flux suggests preferential melting across the year in the MIZ, where both thermal and haline component of the buoyancy flux play a role in the MIZ. In both regions, the buoyancy flux in the leads follow the overall seasonal cycle of the buoyancy fluxes underneath the sea ice floes. The buoyancy flux of leads in the MIZ explain $0.003m^2/s^3$ and $0.13m^2/s^3$ in winter and summer, respectively. Meanwhile, the buoyancy flux of leads in the ice pack explain $-0.05m^2/s^3$, and $0.07m^2/s^3$ in winter and summer, respectively. In other words, leads explain between 20-50% of the buoyancy fluxes in the MIZ and between 15-20% of the buoyancy fluxes in ice pack over the year. Moreover, the area weighted buoyancy fluxes in figure 3c show that winter leads in the MIZ and the summer leads in the ice pack exhibit larger buoyancy fluxes per unit of area compared to those in the MIZ and ice pack, respectively. The mean winter buoyancy flux of the leads in the MIZ is $-1.98 \times 10^{-14}m^2/s^3$, representing a 40% larger buoyancy flux in winter within the leads compared to those in the $(-1.20 \times 10^{-14}m^2/s^3)$, suggesting more freezing within the leads compared to

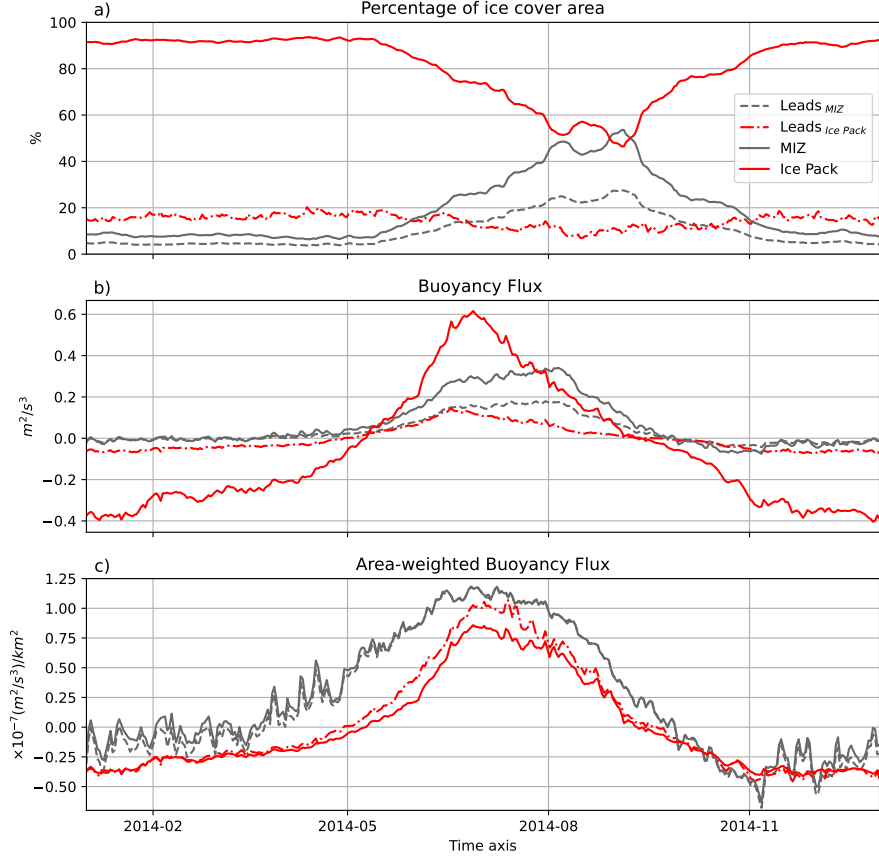


Fig. 3 Time-series during 2014 of a) the area cover, b) area-weighted buoyancy fluxes, c) area-weighted thermal buoyancy flux component for linear kinematic features in the ice pack, the linear kinematic features in the marginal ice zone, the marginal ice zone (ice concentration 0-80%), and the ice pack (ice concentration > 80%).

underneath the floes. The mean summer buoyancy flux of leads in the ice pack is $7.26 \times 10^{-14} \text{ m}^2/\text{s}^3$, approximately 20% more buoyancy flux in summer compared to the ice pack ($5.92 \times 10^{-14} \text{ m}^2/\text{s}^3$), indicating enhanced melting with in the ice pack leads.

To further quantify the role and impact of leads in the Arctic circulation, we use the surface water mass transformation framework (WMT; Walin, 1982). This approach provides information on the transport of water through the surface density contours due to surface buoyancy forcing that results in the formation of lighter or denser water at the surface. In figure 4a, the averaged WMT during 2014 is shown for the ice covered Arctic, the ice pack, the MIZ, and the leads for each of these areas. The yearly mean WMT in the ice pack is characterized by a broad region of strong positive transformation (losing buoyancy) in halocline polar surface water (PSW_H ; $24.36 \text{ kg}/\text{m}^3 < \rho < 27.41 \text{ kg}/\text{m}^3$), reaching a maximum of $\sim 2.1 \text{ Sv}$. Additionally, there

323 is a small negative transformation (gaining buoyancy) below $\rho < 23$. The yearly mean
 324 WMT in the ice pack leads also features a positive peak within the PSW_H reaching
 325 up to 0.39 Sv. On average the WMT in the ice pack leads accounts for $\sim 10\%$ of the
 326 WMT of the ice pack. The MIZ mean WMT is mostly negative within the density
 327 range of $20 < \rho < 27.5$ (gaining buoyancy), reaching a minimum of -1.2 Sv around
 328 $\rho = 27$. The MIZ leads WMT is also negative within the same range of densities,
 329 with a minimum of -0.6 Sv. On average, the WMT in the MIZ leads corresponds to
 330 $\sim 50\%$ of the WMT of the MIZ. Additionally, the sum of the WMT of the ice pack
 331 and the WMT of the MIZ, is in agreement with the surface Arctic WMT reported
 332 by Pemberton et al (2015). The transformation of PSW_H occurs primarily within the
 333 Eurasian basin and a smaller contribution in the Chukchi sea. The transformation of
 334 PSW_{ML} ($\rho < 24.36\text{kg/m}^3$) occurs in the Canadian basin, and the PDW and AW
 335 ($27.41\text{kg/m}^3 < \rho < 27.81\text{kg/m}^3$ and $27.81\text{kg/m}^3 < \rho$, respectively) occurs in the
 336 Barents and Kara seas (Fig.SS3).

337 Hovmoller diagrams of the seasonal cycle of the WMT during 2014 for the ice pack,
 338 the MIZ, and leads in each of these regions are depicted in figure 4. Overall, in the
 339 ice pack and MIZ, winter months show a positive transformation (loss of buoyancy),
 340 due to brine rejection during sea ice growth. Meanwhile, summer presents negative
 341 transformation or a gain of buoyancy occurs due to the melting of sea ice. The mean
 342 WMT underneath floes in the ice pack varies from 1.32 Sv in winter and -2.38 Sv in
 343 summer, and the WMT in the leads of the ice pack is 0.21 Sv in winter and -0.85 Sv
 344 in summer. In other words, the WMT in the ice pack leads explain between 15 to 35%
 345 of the WMT over the year. Furthermore, it is important to note that there is not a
 346 emergence of a pattern in the WMT ratios suggesting that LKFs transform the same
 347 water masses than those transformed underneath the ice pack floes (4 g). The WMT
 348 in MIZ leads in summer is comparable to the summer MIZ WMT. In winter MIZ leads
 349 transform -0.58 Sv, approximately 50% of the winter MIZ WMT (-1.1 Sv). Similarly
 350 to the leads in the ice pack, there is no emergence of a pattern in WMT ratios, thus
 351 suggesting that the MIZ leads WMT is comparable to the MIZ WMT. Overall, the
 352 ice covered Arctic, the sum of the ice pack and the MIZ has a mean positive WMT
 353 during winter of 1.37 Sv and mean negative WMT of -2.38 Sv during summer (4 b).
 354 The ice covered lead WMT is 0.27 Sv in winter and -0.85 Sv in summer (4 c). Thus,
 355 leads in the ice covered Arctic contribute between 20-35% of the WMT throughout
 356 the year, that corresponds to the area covered by the leads in the ice covered Arctic.

357 In figure 4k, the formation rate integrated in 0.5kg/m^3 density bins for the different
 358 ice covered masks, and the leads within each area. The formation rate is the derivative
 359 of the mean WMT with respect to the density and it corresponds to the convergence
 360 of volume in density space. A negative formation in the upper ocean over a density
 361 range indicates that water must be upwelled from the ocean interior or destructed into
 362 a different water mass. We note that there is an agreement in the sign of the water
 363 mass formation for each region, this is consistent with the formation of similar water
 364 masses in leads and underneath floes. There is a notable destruction of water in the
 365 PDW ($27.41\text{kg/m}^3 < \rho < 27.81\text{kg/m}^3$) dominated by the water mass formation in
 366 the MIZ and the MIZ leads. This is likely a consequence of ice freezing and melting
 367 making a dominant contribution to the destruction in the regions where the PDW
 368

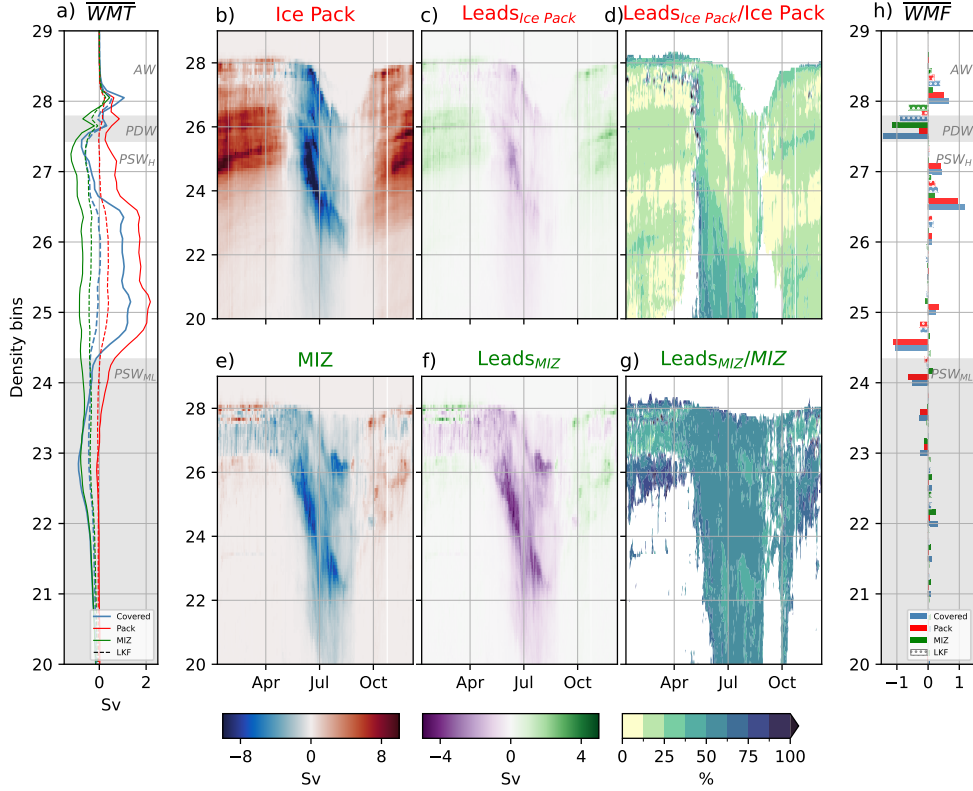


Fig. 4 Mean and daily Water mass transformation over 2014. a) Mean water mass transformation over 2014 for the ice pack, MIZ, and leads in each region. Hovmöller diagram of the water mass transformation (Sv) for the b) ice pack, c) ice pack linear kinematic features and d) the ratio of water mass transformation between the leads and the ice pack region. Panels e), f), and g) follow the same structure as panels b), c) and d), but for the MIZ, MIZ leads, and ratio between them. h) Mean water mass formation in density bins of 0.5kg/m^3 during 2014 for the same regions as in panel a). Shaded gray and white areas in panels a) and h) represent the water masses of the Arctic ocean (Pemberton et al, 2015): the mixed layer polar surface water (PSW_{ML} ; $\rho < 24.36\text{kg/m}^3$), the halocline polar surface water (PSW_H ; $24.36\text{kg/m}^3 < \rho < 27.41\text{kg/m}^3$), the polar deep water (PDW ; $27.41\text{kg/m}^3 < \rho < 27.81\text{kg/m}^3$), and the Atlantic water (AW ; $27.81\text{kg/m}^3 < \rho$).

outcrops within the MIZ. In contrast, AW ($27.81\text{kg/m}^3 < \rho$) and part of the PSW_H ($26\text{kg/m}^3 < \rho < 27.41\text{kg/m}^3$) formation occurs principally in the ice pack, with little contribution from the leads. Finally, the formation/destruction of other water masses (e.g., PSW_{ML} and lower branch of PSW_H) occur predominantly underneath floes.

4 Conclusions

Linear kinematic features are ubiquitous in the sea ice covered oceans and they allow exchanges between the ocean and the atmosphere. Generally, the importance of these fluxes at the ocean-atmosphere interface are considered to be of significant importance

415 for the Arctic dynamics (Maykut and McPhee, 1995). The present study assesses
 416 the impact of leads in the ocean surface by diagnosing the buoyancy fluxes using
 417 high-resolution hindcast (SEDNA). The lead identification algorithm implemented in
 418 SEDNA allows us to reproduce the leads spatial distribution over most regions of
 419 the Arctic, with exception of the Beaufort Sea. Despite these regional differences,
 420 SEDNA can estimate the mean lead probability, showcasing its capacity to capture
 421 the dynamics of these features. This high resolution simulation allows us to better
 422 understand the role of leads in shaping the Arctic ocean.

423 Using SEDNA, we found that leads cover $\sim 20\%$ of the sea-ice covered Arctic, and
 424 in the ice pack, leads can contribute up to 20% more buoyancy flux compared to the
 425 fluxes underneath floes. Despite these larger buoyancy fluxes, leads only account for
 426 approximately $\sim 20\%$ of the total surface water mass transformation in the ice covered
 427 Arctic, equivalent to their area coverage. Thus, contrary to previous hypotheses, our
 428 findings suggest that leads have a small contribution to the transformation of surface
 429 water masses and the Arctic Ocean dynamics than previously thought (e.g. Hutter
 430 et al, 2019; McPhee et al, 2005). While their presence can induce localized changes
 431 in the density of surface waters due to brine rejection and ice melting, their spatial
 432 extent and duration are not sufficient to significantly alter the water masses and
 433 link the surface with the ocean interior. Furthermore, both the transformation and
 434 formation of water masses in the leads have the same sign, and they are comparable to
 435 those underneath the surrounding ice floes. Thus, there is no evidence that leads act
 436 differently or that they have a particular imprint in the Arctic ocean water masses.

437 As discussed by McPhee et al 2005, leads generated by shear can significantly influ-
 438 ence in the ocean mixed layer depth (MLD), by upwelling the isopycnals through
 439 Ekman pumping linked to shear in the ice-ocean stress at the location of the leads.
 440 Although the MLD is shallower in the leads than underneath the flows in SEDNA (Fig.
 441 SS2). For example, the difference in MLD between the ice pack and ice pack leads is
 442 significant and, on average, the MLD is $\sim 20\%$ shallower during winter in the ice pack
 443 leads. Nevertheless, there is no evidence of an impact in the ice cover nor a distinct
 444 response to the surface buoyancy fluxes or change of the water mass transformation
 445 within the leads. Thus, it remains to be investigated if the impact of entrainment in
 446 the MLD within the leads, as well as the source of potential energy and generation
 447 of MLD instabilities due to isopycnal tilting in the vicinity of leads could impact the
 448 Arctic ocean dynamics.

449 Leads have an important local effect in the atmosphere and the sea-ice fluxes
 450 (Lüpkes et al, 2008; Marcq and Weiss, 2012; Wettlaufer et al, 1997; Smith et al, 1990),
 451 but a limited imprint in the ocean fluxes. Our results suggest that the representation
 452 of leads in oceanic climate models is not priority, since there is no evidence in our
 453 high-resolution simulation that the fluxes through leads will impact significantly the
 454 stratification or the dynamics of the Arctic ocean. Furthermore, leads are likely to
 455 have a negligible impact improving the well known issue within the Arctic climate
 456 modeling community of the misrepresentation of the stratification and water masses
 457 in the Arctic ocean (Wang et al, 2024; Ilıcak et al, 2016). Finally, a limitation of
 458 this study is the use of a visco-plastic rheology that results in leads with high ice
 459 thickness and concentrations, thus limiting the exchanges between the ocean and the
 460

atmosphere, as compare to the elasto-brittle rheology that may allow larger fluxes through the leads (Rampal et al, 2016; Ólason et al, 2021).

5 Methods

Model description

Here we employ the NEMO numerical platform (release 4.0.5) to establish a 1/60° pan-Arctic ocean-sea ice configuration named SEDNA. The ocean core (NEMO-OCE) solves the primitive equations (PE) using finite differences on an Arakawa C-grid mesh, while the ocean is coupled to the SI3 sea-ice component (NEMO-SI3). The pan-Arctic domain covers the Bering Strait on the Pacific side and extends southward to 56°N in the Subpolar North Atlantic gyre and to the Baltic Sea entrance. The ocean model incorporates a linear free surface, utilizes a 3rd order flux form scheme for momentum advection, and employs a 4th order Flux Corrected Transport (FCT) for tracer advection. Additionally, the sea-ice representation includes a 5-category model with the Elasto-Visco-Plastic (EVP) rheology activated and a land-fast capability to better simulate sea-ice behavior in specific areas.

The NCAR bulk formula from Large & Yeager (2009) to calculate air-sea fluxes and uses hourly ERA5 atmospheric data for near-surface variables. The three lateral boundaries are constrained with daily GLORYS12V1 Re-analysis data, and monthly freshwater fluxes from river discharges are combined with Greenland land ice melt data to supply SEDNA. The simulation extends over seven years (2009 to 2015) and starts from rest, utilizing initial conditions based on the World Ocean Atlas 2009 temperature/salinity and mean January 2009 ice state from PIOMAS re-analysis. Preliminary assessment against available observations and re-analyses data shows that the simulation reasonably captures key ocean and sea-ice characteristics.

Identification of leads

The leads from SEDNA were identified by using the algorithm proposed by Hutter et al (2019). This method uses the total deformation rate defined as:

$$T_d = \left[\underbrace{\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)^2}_{Divergence} + \underbrace{\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2}_{Shear} \right]^{1/2}, \quad (1)$$

where u and v are the sea ice velocities. The deformation rate varies spatially depending on the background deformation and the sea ice properties (Reiser et al, 2020), where the largest deformation rates occur along the boundaries of ice flows where the leads are found. Therefore, to identify the leads, the local maxima is identified using a DoG filter in the deformation rate field. After this, a binary map is created with the mask of identified lead. Note that after this step, Hutter et al (2019) reduce the leads width to a line to then track the leads over time. Since our study focus on the ocean-atmosphere interactions, we only use the identification algorithm to produce a mask

of leads during 2014 for the Arctic, where we retain the original width computed from the deformation rate. Finally, the lead mask is then used to quantify and contrast the impact of leads compared to the ice covered Arctic, the ice-pack (ice concentration > 80%) and the MIZ (0 - 80% ice concentration).

Buoyancy flux and water mass transformation

The Arctic ocean surface experiences large heat and freshwater fluxes, because during the formation of sea ice, salt is rejected and the surface cooled, and when ice melts, freshwater is returned to the ocean and the surface warmed. These processes change significantly the surface buoyancy making the surface denser when salt is rejected and lighter when freshwater is added at the surface. This seasonal melt-freeze cycle changes buoyancy and transforms the density of the surface water masses (water mass transformation). Walin (1982) proposed the water mass transformation framework that quantifies the rate at which a given water mass is made lighter (negative transformation rate) or denser (positive) due to surface fluxes and mixing.

The surface water mass transformation is computed as:

$$\Omega(\sigma_k, t) = - \underbrace{\frac{1}{\sigma_{k+1} - \sigma_k} \iint_A \left(\frac{\alpha Q_{net}}{\rho_0 C_p} \right) dA}_{Net\ surface\ heat\ flux} + \underbrace{\frac{1}{\sigma_{k+1} - \sigma_k} \iint_A \left(\frac{\beta F_{net} SSS}{\rho_0} \right) dA}_{Net\ surface\ freshwater\ flux} \quad (2)$$

where σ is potential density, t is time, Q_{net} is the net heat flux at the ocean surface, C_p is the specific heat capacity of sea-water, β is the haline contraction, F_{net} is the net freshwater flux at the ocean surface and SSS the sea surface salinity. The transformation rate ω was spatially decomposed into different sea-ice dynamics such as the ice covered, the pack ice, the MIZ, and the leads in each of these regions. By separately diagnosing the density transformation that occurs in the leads and underneath the surrounding floes, we quantify the relative importance of the leads for each of these regions. Additionally, this framework accounts for the area changes of the ice cover during the seasonal cycle, and excludes open ocean regions.

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References

von Albedyll L, Hendricks S, Grodofzig R, et al (2022) Thermodynamic and dynamic contributions to seasonal Arctic sea ice thickness distributions from airborne observations. *Elementa: Science of the Anthropocene* 10(1). <https://doi.org/10.1525/elementa.2021.00074>

Bourgault P, Straub D, Duquette K, et al (2020) Vertical Heat Fluxes beneath Idealized Sea Ice Leads in Large-Eddy Simulations: Comparison with Observations from the SHEBA Experiment. <i>Journal of Physical Oceanography</i> 50(8):2189–2202. https://doi.org/10.1175/jpo-d-19-0298.1	553 554 555 556 557
Hutchings JK, Heil P, Hibler WD (2005) Modeling Linear Kinematic Features in Sea Ice. <i>Monthly Weather Review</i> 133(12):3481–3497. https://doi.org/10.1175/mwr3045.1	558 559 560 561
Hutter N, Losch M, Menemenlis D (2018) Scaling Properties of Arctic Sea Ice Deformation in a High-Resolution Viscous-Plastic Sea Ice Model and in Satellite Observations. <i>Journal of Geophysical Research: Oceans</i> 123(1):672–687. https://doi.org/10.1002/2017jc013119	562 563 564 565 566
Hutter N, Zampieri L, Losch M (2019) Leads and ridges in Arctic sea ice from RGPS data and a new tracking algorithm. <i>The Cryosphere</i> 13(2):627–645. https://doi.org/10.5194/tc-13-627-2019	567 568 569 570
Ilicak M, Drange H, Wang Q, et al (2016) An assessment of the arctic ocean in a suite of interannual core-ii simulations. part iii: Hydrography and fluxes. <i>Ocean Modelling</i> 100:141–161. https://doi.org/https://doi.org/10.1016/j.ocemod.2016.02.004 , URL https://www.sciencedirect.com/science/article/pii/S1463500316000238	571 572 573 574
Linow S, Dierking W (2017) Object-Based Detection of Linear Kinematic Features in Sea Ice. <i>Remote Sensing</i> 9(5):493. https://doi.org/10.3390/rs9050493	575 576 577
Lüpkes C, Vihma T, Birnbaum G, et al (2008) Influence of leads in sea ice on the temperature of the atmospheric boundary layer during polar night. <i>Geophysical Research Letters</i> 35(3). https://doi.org/10.1029/2007gl032461	578 579 580 581
Marcq S, Weiss J (2012) Influence of sea ice lead-width distribution on turbulent heat transfer between the ocean and the atmosphere. <i>The Cryosphere</i> 6(1):143–156. https://doi.org/10.5194/tc-6-143-2012	582 583 584 585
Maykut GA (1986) <i>The Surface Heat and Mass Balance</i> , Springer US, Boston, MA, pp 395–463. https://doi.org/10.1007/978-1-4899-5352-0_6	586 587 588
Maykut GA, McPhee MG (1995) Solar heating of the Arctic mixed layer. <i>Journal of Geophysical Research: Oceans</i> 100(C12):24691–24703. https://doi.org/10.1029/95jc02554	589 590 591
McPhee MG, Kwok R, Robins R, et al (2005) Upwelling of arctic pycnocline associated with shear motion of sea ice. <i>Geophysical Research Letters</i> 32(10). https://doi.org/10.1029/2004GL021819	592 593 594 595
Pemberton P, Nilsson J, Hieronymus M, et al (2015) Arctic Ocean Water Mass Transformation in S–T Coordinates. <i>Journal of Physical Oceanography</i> 45(4):1025–1050.	596 597 598

599 <https://doi.org/10.1175/jpo-d-14-0197.1>
600
601 Rampal P, Bouillon S, Ólason E, et al (2016) neXtSIM: a new Lagrangian sea ice
602 model. *The Cryosphere* 10(3):1055–1073. <https://doi.org/10.5194/tc-10-1055-2016>
603
604 Reiser F, Willmes S, Heinemann G (2020) A New Algorithm for Daily Sea Ice Lead
605 Identification in the Arctic and Antarctic Winter from Thermal-Infrared Satellite
606 Imagery. *Remote Sensing* 12(12):1957. <https://doi.org/10.3390/rs12121957>
607
608 Richter-Menge JA, McNutt SL, Overland JE, et al (2002) Relating arctic pack ice
609 stress and deformation under winter conditions. *Journal of Geophysical Research:*
610 *Oceans* 107(C10):SHE 15–1–SHE 15–13. <https://doi.org/10.1029/2000jc000477>
611
612 Ruffieux D, Persson POG, Fairall CW, et al (1995) Ice pack and lead surface energy
613 budgets during leadex 1992. *Journal of Geophysical Research: Oceans* 100(C3):4593–
614 4612. <https://doi.org/10.1029/94JC02485>
615
616 Smith DC, Morison JH (1998) Nonhydrostatic haline convection under leads in sea
617 ice. *Journal of Geophysical Research: Oceans* 103(C2):3233–3247. <https://doi.org/10.1029/97jc02262>
618
619 Smith DC, Lavelle JW, Fernando HJS (2002) Arctic Ocean mixed-layer eddy
620 generation under leads in sea ice. *Journal of Geophysical Research: Oceans*
621 107(C8):17–1–17–17. <https://doi.org/10.1029/2001jc000822>
622
623 Smith SD, Muench RD, Pease CH (1990) Polynyas and leads: An overview of physical
624 processes and environment. *Journal of Geophysical Research: Oceans* 95(C6):9461–
625 9479. <https://doi.org/10.1029/jc095ic06p09461>
626
627 Talandier C, Lique C (2023) Sedna-delta. <https://doi.org/10.5281/zenodo.7656924>,
628 URL <https://doi.org/10.5281/zenodo.7656924>
629
630 Tschudi M, Curry J, Maslanik J (1998) Airborne observations of leads in the Beaufort
631 Sea. *IGARSS '98 Sensing and Managing the Environment 1998 IEEE Interna-*
632 *tional Geoscience and Remote Sensing Symposium Proceedings (Cat No98CH36174)*
633 2:986–988 vol.2. <https://doi.org/10.1109/igarss.1998.699648>
634
635 Untersteiner N (1961) On the mass and heat budget of arctic sea ice. *Archiv für*
636 *Meteorologie, Geophysik und Bioklimatologie, Serie A* 12(2):151–182. <https://doi.org/10.1007/bf02247491>
637
638 Walin G (1982) On the relation between sea-surface heat flow and thermal circula-
639 tion in the ocean. *Tellus* 34(2):187–195. <https://doi.org/10.1111/j.2153-3490.1982.tb01806.x>
640
641
642 Wang Q, Danilov S, Jung T, et al (2016) Sea ice leads in the Arctic Ocean:
643 Model assessment, interannual variability and trends. *Geophysical Research Letters*
644

43(13):7019–7027. https://doi.org/10.1002/2016gl068696	645
Wang Q, Shu Q, Bozec A, et al (2024) Impact of increased resolution on arctic ocean simulations in ocean model intercomparison project phase 2 (omip-2). <i>Geoscientific Model Development</i> 17(1):347–379. https://doi.org/10.5194/gmd-17-347-2024 , URL https://gmd.copernicus.org/articles/17/347/2024/	646 647 648 649 650 651
Weiss J, Marsan D (2004) Scale properties of sea ice deformation and fracturing. <i>Comptes Rendus Physique</i> 5(7):735–751. https://doi.org/10.1016/j.crhy.2004.09.005	652 653 654 655
Wernecke A, Kaleschke L (2015) Lead detection in Arctic sea ice from CryoSat-2: quality assessment, lead area fraction and width distribution. <i>The Cryosphere</i> 9(5):1955–1968. https://doi.org/10.5194/tc-9-1955-2015	656 657 658 659
Wettlaufer JS, Worster MG, Huppert HE (1997) The phase evolution of Young Sea Ice. <i>Geophysical Research Letters</i> 24(10):1251–1254. https://doi.org/10.1029/97gl00877	660 661 662
Willmes S, Heinemann G, Schnaase F (2023) Patterns of wintertime Arctic sea ice leads and their relation to winds and ocean currents. <i>The Cryosphere Discussions</i> 2023:1–23. https://doi.org/10.5194/tc-2023-22	663 664 665 666
Ólason E, Rampal P, Dansereau V (2021) On the statistical properties of sea-ice lead fraction and heat fluxes in the Arctic. <i>The Cryosphere</i> 15(2):1053–1064. https://doi.org/10.5194/tc-15-1053-2021	667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690

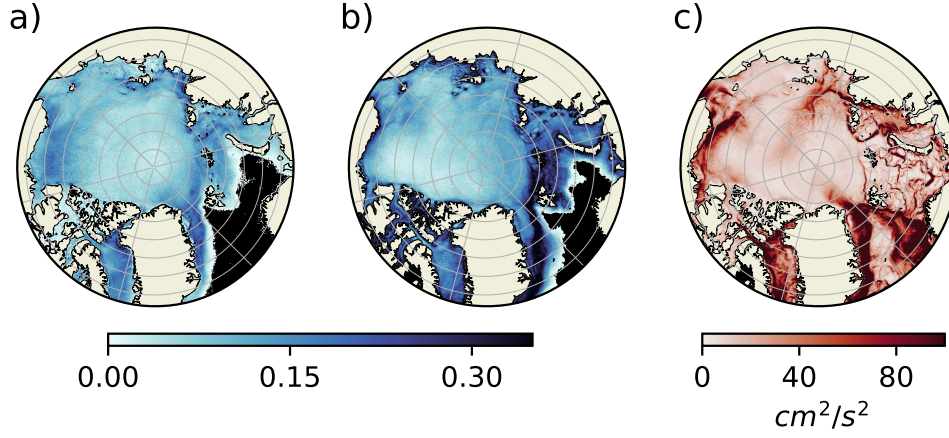


Fig. S1 Linear kinematic mean frequency from a) Thermal-Infrared Satellite Imagery (Willmes et al, 2023) and b) SEDNA between 2011 and 2015. Panel c) shows the mean kinetic energy between 2011 and 2015.

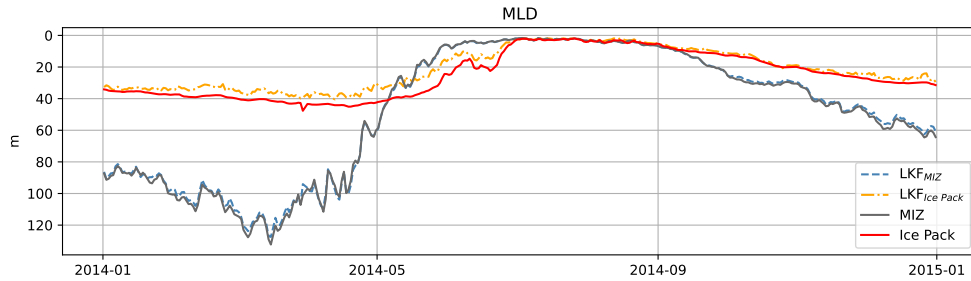


Fig. S2 Time-series during 2014 of the mean mixed layer depth for linear kinematic features in the ice pack, the linear kinematic features in the marginal ice zone, the marginal ice zone (ice concentration 0-80%), and the ice pack (ice concentration > 80%). The mixed layer depth is referenced to the first ocean model level.

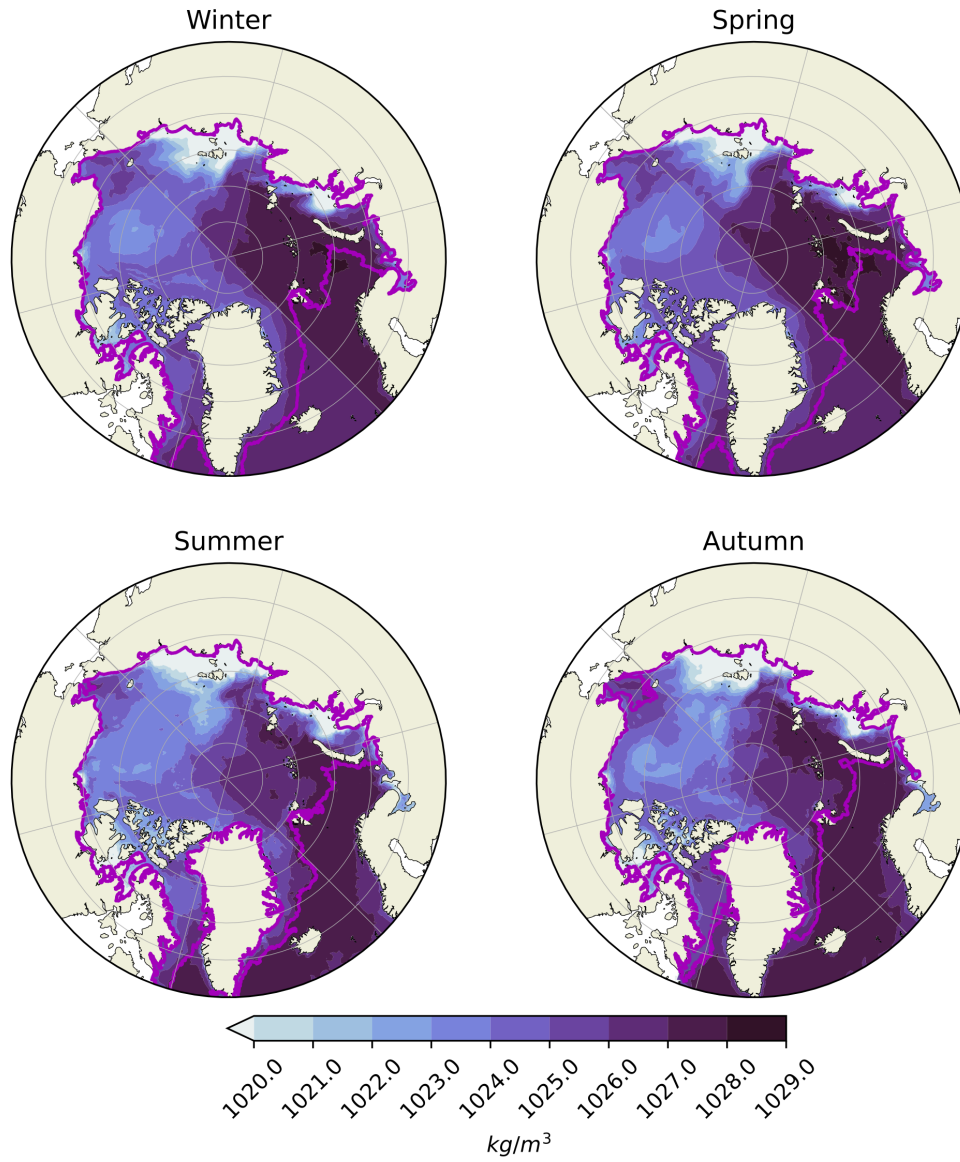


Fig. S3 Maps of the seasonal surface density of SEDNA during 2014. Magenta contour in maps correspond to the 0% concentration contour of sea ice.